

INTRODUCTION TO DATABASE

Class: B2EM

Final examination

Duration: 1h30min

Part 2: Practice

The following practical part contains 12 questions with 12 marks available.

1. Copy and open the workfile called **library.accdb** to your folder. Open the **Member** table.
2. Use Design View to add fields to the **Member** table as indicated:

- **gender**: Text field of length 6 to accommodate the values male and female. Make this a required field that is not indexed
- **phoneNumber**: a number of size 9.

Switch back to Datasheet View (You must reply yes to the system prompt to save your changes). You should notice there are no values for gender nor birthDate.

3. Now enter values as shown in the table below in the gender and birthdate fields for each member

firstName	lastName	gender	phoneNumber
John	Smith	Male	655345421
David	Martin	Male	678421444
Betty	Freeman	Female	685421354
John	Martin	Male	698754213

4. Add the following records to the **Member** table. Save and close the table

firstName	lastName	gender	phoneNumber
Priso	Pondi	Female	69993214

5. Open the **Book** table and sort the table by **title** in ascending order, then export the **Book** table in Microsoft Excel format as **Book.xlsx** and save to your folder. Save and close the **Book** table.
6. Typically a library assesses a fine the user must pay if they keep a book out past the due date. As well the library needs to track the amount, if any, the member has paid. a) Open the **Loan** table in design view and add two new fields named **fineAssessed** and **finePaid**. These fields must have a datatype of **Currency**. b) Save the **Loan** table and then view the rows of the table. There are no amounts for these fields. c) Choose some row(s) in the **Loan** table and enter values for the **fineAssessed** and **finePaid** fields. Note the values you enter will appear as dollars and cents.

7. In exercise 4 you added a new member and in exercise 6 you added fields to the **Loan** table. Consider that the person you added now borrows a book and so a row must be entered into the **Loan** table. Enter such a row.

8. Consider the **Member** table. Previously you added a **gender** field. Open the **Member** table in design view and change the datatype for gender to be **Lookup Wizard**. The wizard will automatically present 3 successive popup windows where you will:

- i) specify that you are providing the lookup values;
- ii) enter the values (Male and Female);
- iii) specify that values are to be limited to your list.

Save the table and enter datasheet view so you can test out the datatype you have just created. You will notice the user sees a drop down list containing Male and Female, and so the user cannot enter/select an inappropriate value.

9. Consider the **gender** and **phoneNumber** fields in the **Member** table of the **Library** database. Modify the caption (légende) for these fields to be **Gender** and **Phone Number** respectively. Save the table and reopen in datasheet view. You will see these captions at the top of their respective columns.

10. Consider the **Loan** table and its date fields. MS Access has many built in functions one of which is **Date()** which always returns today's date. So, to ensure that someone always enters a due date later than today, in the properties section for the date due field enter the following:

- Validation rule: **>= Date()**
- Validation text: **Enter a future date.**

11. Create a query to list the number of members by gender.

12. Create a query to list books due on a specific date (a parameter).